

# Multiple Linear Regression

## Parameter Estimation via MLE

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# Lecture Outline

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- ▶ Assessing Model Fit: R-squared and Adjusted R-squared

# Recap: Simple Linear Regression

- ▶ Models the relationship between a single predictor variable,  $x$ , and a response variable,  $y$ .
- ▶ The model is a straight line:  $y \approx \beta_0 + \beta_1 x$ .
- ▶ We found the "best" line by minimizing the Sum of Squared Residuals (SSR), which is equivalent to Maximum Likelihood Estimation (MLE) under the assumption of normally distributed errors.

# A Motivating Example: House Prices Revisited

**The Old Question:** Can we predict the price of a house based on its size?

**A New Question:** What if other factors also influence the price? For instance, what about the number of bedrooms?

# A Motivating Example: House Prices

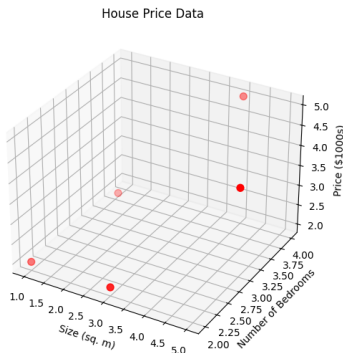
**Scenario:** You are a real estate analyst. You have collected data on 5 recent house sales, recording the size of the house, the number of bedrooms, and its selling price.

**The Core Question:** Can we build a model to predict the price of a house based on its size **and** the number of bedrooms?

| Size (sq. m, $x_1$ ) | Bedrooms ( $x_2$ ) | Price (\$1000s, $y$ ) |
|----------------------|--------------------|-----------------------|
| 1                    | 2                  | 2                     |
| 2                    | 3                  | 3                     |
| 3                    | 2                  | 2                     |
| 4                    | 4                  | 5                     |
| 5                    | 3                  | 4                     |

# Visualizing the Data: A 3D Scatter Plot

With two predictor variables, we can no longer use a simple 2D scatter plot. We now need to visualize the data in three dimensions.



**Figure:** A 3D scatter plot showing the relationship between size, bedrooms, and price.

# The Multiple Linear Regression Model

The simplest way to model this trend is to fit a plane through the data.

**The Model:**  $y \approx \beta_0 + \beta_1 x_1 + \beta_2 x_2$

- ▶  $\beta_0$ : The intercept (the price of a 0 sq. m house with 0 bedrooms)
- ▶  $\beta_1$ : The slope for size (the increase in price for each extra sq. m, holding the number of bedrooms constant)
- ▶  $\beta_2$ : The slope for bedrooms (the increase in price for each extra bedroom, holding the size constant)

This can be generalized to  $p$  predictor variables:

$$y \approx \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p$$

# The Model in Matrix Form

For a more compact notation, we can express the model using matrices.

$$y = X\beta + \epsilon$$

Where:

- ▶  $y$  is the vector of observed outcomes.
- ▶  $X$  is the design matrix, which includes a column of ones for the intercept and a column for each predictor variable.
- ▶  $\beta$  is the vector of coefficients.
- ▶  $\epsilon$  is the vector of errors.

For our house price example:

$$y = \begin{pmatrix} 2 \\ 3 \\ 2 \\ 5 \\ 4 \end{pmatrix}, X = \begin{pmatrix} 1 & 1 & 2 \\ 1 & 2 & 3 \\ 1 & 3 & 2 \\ 1 & 4 & 4 \\ 1 & 5 & 3 \end{pmatrix}, \beta = \begin{pmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{pmatrix}, \epsilon = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \\ \epsilon_4 \\ \epsilon_5 \end{pmatrix}$$



# Geometric Approach: Minimizing Vertical Distance

Similar to simple linear regression, we want to find the plane that is "closest" to our data points. We measure this closeness by the vertical distance from each point to the plane, called the residual.

**We want to minimize:**

$$SSR = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip}))^2$$

In matrix form, this is:

$$SSR = (y - X\beta)^T (y - X\beta)$$

# A More Principled Approach: MLE

The justification for minimizing the sum of squared residuals comes from Maximum Likelihood Estimation (MLE).

**The Linear Regression Model Assumption:** The deviation from the plane (the error) is random noise that follows a Normal (Gaussian) distribution.

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \quad \text{where } \epsilon_i \sim N(0, \sigma^2)$$

- ▶ The mean of the errors is 0 (the plane is correct on average).
- ▶ The errors have a constant variance  $\sigma^2$ .

# Solving for the MLE

Maximizing the log-likelihood for the multiple linear regression model is equivalent to minimizing the Sum of Squared Residuals (SSR).

By taking the derivative of the SSR with respect to  $\beta$  and setting it to zero, we get the **Normal Equations**:

$$(X^T X)\beta = X^T y$$

The solution for the coefficient vector is:

$$\hat{\beta} = (X^T X)^{-1} X^T y$$

# Interpreting the Coefficients

The interpretation of the coefficients in multiple linear regression is slightly different from simple linear regression.

Each coefficient,  $\beta_j$ , represents the expected change in the response variable,  $y$ , for a one-unit change in the predictor variable,  $x_j$ , **while holding all other predictor variables constant.**

# Assessing Model Fit: R-squared

**R-squared** ( $R^2$ ) is the proportion of the variance in the dependent variable that is predictable from the independent variable(s).

$$R^2 = 1 - \frac{SSR}{SST} = 1 - \frac{\sum (y_i - \hat{y}_i)^2}{\sum (y_i - \bar{y})^2}$$

- ▶ A value of 0 indicates that the model explains none of the variability of the response data around its mean.
- ▶ A value of 1 indicates that the model explains all the variability of the response data around its mean.

# A Problem with R-squared

A potential issue with R-squared is that it will always increase when you add more predictor variables to the model, even if those variables are not actually useful.

This can lead to overfitting, where the model performs well on the training data but poorly on new, unseen data.

**Solution:** Adjusted R-squared.

# Adjusted R-squared

Adjusted R-squared is a modified version of R-squared that has been adjusted for the number of predictors in the model.

$$R_{adj}^2 = 1 - (1 - R^2) \frac{n - 1}{n - p - 1}$$

Where:

- ▶  $n$  is the number of observations.
- ▶  $p$  is the number of predictor variables.

Adjusted R-squared only increases if the new term improves the model more than would be expected by chance. It can decrease if a poor-quality predictor is added to the model.

# Why Linear Regression is Still Important

## Advantages

- ▶ **Interpretability:** Very easy to understand and explain.
- ▶ **Simplicity & Speed:** It is computationally inexpensive and fast to train.
- ▶ **Foundation:** It is the basis for many more advanced statistical and machine learning models (e.g., Logistic Regression, GLMs, XAI).

## Applications in AI and Data Science

- ▶ **Baseline Model:** It provides a baseline performance that more complex models (like neural networks) must beat.
- ▶ **Forecasting:** Used in economics and finance to predict variables like GDP, sales, or demand based on time or other factors.
- ▶ **Understanding Relationships:** Used in social sciences and other fields to understand the relationships between multiple variables.



# Summary

- ▶ **Multiple Linear Regression** models the relationship between two or more predictor variables and a response variable.
- ▶ The goal is to find the "best" hyperplane that fits the data.
- ▶ The geometric approach (OLS) and the statistical approach (MLE) both lead to the same solution: minimizing the sum of squared residuals.
- ▶ Coefficients are interpreted as the change in the response for a one-unit change in a predictor, holding all other predictors constant.
- ▶ Adjusted R-squared is a better measure of model fit than R-squared when comparing models with different numbers of predictors.